2.4 Modeling with Quadratic Functions

Name _____ Period ____ Date _____

Algebra II 2.4 · Student sheet

I can…

- write quadratic equations using vertex form and intercept form.
- use finite differences to tell whether data are linear, quadratic, or neither.
- model real-life data and situations with quadratic functions.

Part 1 · Ball toss data

Time (s)	0	0.5	1	1.5	2	2.5
Height (ft)	6	22	30	30	22	6
First differences						
Second differences						

Type of function: _____ When is the ball highest? _____ About how high? _____

Part 2 · Notes: fill in the blanks

Know the vertex: y = _____. Know the x-intercepts: y = _____.

Constant first differences → _____; constant second differences → _____.

Part 3 · Examples (work them with the class)

Example 1

Write an equation of the parabola with vertex (2, 5) that passes through (4, −3).

Example 2

Write an equation of the parabola with x-intercepts −1 and 4 that passes through (2, 12).

Example 3

Tell whether the data are linear, quadratic, or neither. Write a function for the data.

x	0	1	2	3	4	5
y	2	5	12	23	38	57

Example 4

You have 80 feet of fencing to make a rectangular pen against a barn (the barn is one side). What dimensions give the greatest area? What is that area?

Part 4 · Partner practice

Partner 1

Write an equation of the parabola with vertex $(-3, 1)$ that passes through $(-1, 9)$.

Partner 2

Write an equation of the parabola with x-intercepts 0 and 6 that passes through (3, 18).